MEP Commissioning Standard (Summary)

Stages, witness points and acceptance criteria for mechanical, electrical and plumbing systems

Commissioning proves that a building performs as designed before the owner accepts it. This summary condenses the practice recommended by the Construction & MEP Council into six stages and a set of witness points that the owner's representative should attend.

1. Stages

1. Design review: confirm the owner's project requirements are traceable to the drawings and the control sequences.
2. Submittals and factory tests: witness the chiller, transformer, DG and fire pump tests at the factory where possible.
3. Installation checks: pre-functional checklists for every item, signed by the installer before power is applied.
4. Start-up: manufacturer start-up with the commissioning agent, recording baseline readings.
5. Functional performance tests: run each system through its sequences, failure modes and load steps.
6. Integrated systems test: power failure, fire alarm interface, lift recall, smoke control and BMS response together.

2. Acceptance criteria

System	Test	Pass criterion
Chilled water	8-hour load test at design flow	Supply temp within 0.5 deg C of setpoint; delta-T >= 5.5 deg C
Air side	Air balancing at all terminals	Within +/-10% of design airflow
Electrical	Thermography at full load, protection relay tests	No hot spot over 15 deg C above ambient
DG and ATS	Black-start with building load, 4-hour run	Transfer under 10 seconds; stable frequency
Fire pumps	Flow test at 100% and 150% duty	Pressure and flow per the pump curve
BMS	Point-to-point check, alarm simulation	100% of points verified; alarms reach operators

3. Documentation

- A commissioning plan agreed before the first installation check
- Issues log with severity, owner, closure date and re-test evidence
- Final commissioning report with trend data, test sheets and outstanding items
- Seasonal re-tests scheduled for the first summer and the first monsoon